SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I Lakshumipriya OS (192521197) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

O.S. Lakshumipriya.

Signature of the student:

Annexure A

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security
A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message
A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

a. Deduce the most likely original message.
b. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System
A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

a. Demonstrate how the encryption process transforms the message step-by-step.
b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation
During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

a. Analyze how investigators could detect the presence of hidden data within the image.
b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication
A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".

a. Determine the most probable original message by reversing the encryption process.
b. Critically evaluate why such a method is vulnerable in modern communication systems.
c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools

CNS · Assessment.

CO1 - AT1.

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(1).

(a). Encryption using shift of 4 positions.

* using caesar cipher.

text : MEET AT NOON.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18

T	U	V	W	X	Y	Z
19	20	21	22	23	24	25.

* Encryption : $C = (p+k) \bmod 26$.

text	Value	$(p+k) \bmod 26$	text
M	12	16	Q
E	4	8	I
E	4	8	I
T	19	23	X
A	0	4	E
T	19	23	X
N	13	17	R
O	14	18	S
O	14	18	S
N	13	17	R.

b). Decryption:

→ The authorized member shifts every encrypted letter four positions backward to get the orginal message.

For example, Q → M, I → E, X → T.

the message – MEET AT NOON.

$$P = (C - k) \bmod 26$$

(a) GSRH RH Z HVXIVG.

Plaintext: A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
Cipher: Z Y X W V U T S R Q P O N M L K J I H G F E D C B A.

* Word 1 → GSRH → THIS.
* RH → IS
* Z → A
* HVXIVG – SECRET.

THIS IS A SECRET.

Justification:

* In Atbash cipher, A is replaced by Z
  B is Replaced by Y.
  C by X ... so on

The text of cipher finally becomes

THIS IS A SECRET which is a meaningful english text.

* Letter substitution.
* word structure.
* consistent mapping.
* No Numerical key is needed.

(3) Secure messaging system.

(a). Encryption HELLO → keyword KEY.

plain text →	H	E	L	L	O
	7	4	11	11	14
key word →	K	E	Y	K	E
	10	4	24	10	4
	17	8	9	21	18
	R	I	J	V	S

Encrypted message - RIJVS.

(b) Effect of repeating keyword.

* The keyword KEY is repeated until it matches the length of message.
* Each keyword letter determines the shift applied to corresponding plaintext letter.

For example,

H shifted by K gives R.

L shifted by E gives I.

→ continuous for 5 letter.

• This method uses different shifts instead of single fixed shift, making it more complex than basic caesar cipher.

④

(a) Detecting hidden data in image.

(1) File analysis : Examine the image's metadata, file size, headers and unusual data

(2) Pixel analysis : Inspect pixel values of LSB.

(3) statistical analysis : Look for unusual pattern in pixel distributions that may indicate data.

(4) comparison : compare sus image with orginal copy to identify unexpected changes.

(b). Two techniques to strengthen security.

(1) Encryption before embedding : Encrypt the confidential message using strong encryption algorithm before hiding it in the image. This protects the content even if it is extracted.

(2).

LSB steganography with secure embedding.

→ Distributes encrypted data carefully across suitable image pixels to make it less noticeable.

→ This can reduce simple detection, although it cannot guarantee that the hidden data will remain undetected.

(5)

Given: WECRLTEERDOEEFEOCAIVDEN.

→ This is Transposition cipher, letters are rearranged rather than replaced.

→ A likely interpretation is the three-Rail-Rail fence cipher, a transposition technique that rearranges character without changing the characters themselves.

→ There is problem with the question, the supplied ciphertext does not match the standard three-rail Rail fence encryption of plaintext.

• For the plaintext

WEAREDISCOVERED E LEE AT ONCE.

-It produces,

WECRLTEERDSOEEFEAOCAIVDEN.

with space inserted:

WE ARE DISCOVERED FLEE AT ONCE.

(b).

* No character substitution: The orginal letters remain unchanged; only their positions are rearranged.

+ Limited key space: The no of possible arrangements is small for simple transposition methods.

* Frequency patterns remain: Letter frequencies are preserved, which can help analyze text easily.

*. Easy to break: Attackers can try different arrangement to recover the orginal message

* No modern security protection: It does not provide strong protection.

* combine transposition with strong encryption.

- Encrypt the plaintext using modern authenticated encryption algorithm, such as

* AES-GCM.